MAT3500 - Topology:
Mandatory assignment

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Problem 1

Let (X, d) be a metric space.

(a) Show that $d\colon X \times X \to \mathbb{R}$ is continuous, where $X \times X$ has the product topology.

Proof. Let $B := (a, b)$ be a basis element in $\mathbb{R}$, and $(x, y) \in d^{-1}(B)$.
Then let $\varepsilon := \min\left(\frac{b-d(x,y)}{2}, \frac{d(x,y)-a}{2}\right)$.
Let $U := B_d(x, \varepsilon) \times B_d(y, \varepsilon)$ be an open neighbourhood around (x, y). Let $(x', y') \in U$.
Then by applying the triangle inequality twice, we get:

$$d(x', y') \leq d(x, y) + d(x', x) + d(y, y') < d(x, y) + 2\varepsilon \leq b$$

And by applying the inverse triangle inequality twice we get:

$$d(x', y') \geq d(x, y) - d(x', x) - d(y, y') > d(x, y) - 2\varepsilon \geq a$$

Therefore $(x', y') \in d^{-1}(B)$, which implies $U \subseteq d^{-1}(B)$. Thus, $d^{-1}(B)$ is open.
Then we have that d is continuous. ∎

(b) Show that for $x, y \in X$, subset $A := \{z \in X : d(x, z) = d(y, z)\}$ is closed in X.

Proof. Let $f\colon X \to \mathbb{R}$ be defined by $z \mapsto d(x, z) - d(y, z)$, then we have that $A = f^{-1}(\{0\})$. Since d is continuous it is also continuous in each variable separately, then f is continuous. And since $\{0\}$ is closed, then so is A. ∎

Problem 2

Let $S := \mathbb{N} \cup \{\infty\}$ and X be a topological space. Find a topology on S for which a function $f\colon S \to X$ is continuous if and only if the sequence $f(n)$ converges to $f(\infty)$ in X.

Proof. Let S have the topology generated by the subbasis $\{\{n\}\}_{n\in\mathbb{N}} \cup \{\{n \in S : n \geq N\}\}_{N\in\mathbb{N}}$.

First, assume that f is continuous. Let $U \subseteq X$ be an open neighborhood of $f(\infty)$. Since f is continuous, there exists a basis element B such that $\infty \in B$ and $f(B) \subseteq U$. Since $\infty \in B$, B is a finite intersection of subbasis elements of the form $\{n \in S : n \geq N_i\}$. Thus, $B = \{n \in S : n \geq N\}$ where $N = \max(N_i)$. Therefore, $f(n) \in U$ for all $n \geq N$. Then $f(n) \to f(\infty)$.

Conversely, assume that $f(n) \to f(\infty)$ as $n \to \infty$. Let U be open in X, and let $s \in f^{-1}(U)$. If $s \neq \infty$, then the singleton $\{s\}$ is open, and $s \in \{s\} \subseteq f^{-1}(U)$. If $s = \infty$, by our assumption there exists an $N \in \mathbb{N}$ such that $f(n) \in U$ for all $n \geq N$. Then the open set $\{n \in S : n \geq N\} \subseteq f^{-1}(U)$. Thus, $f^{-1}(U)$ is open, and f is continuous.

■

Problem 3

Let X, Y be topological spaces, $f, g\colon X \to Y$ two continuous functions, and assume that $f|_A = g|_A$ for a dense subset $A \subseteq X$.

(a) Show that if Y is Hausdorff, then $f = g$.

Proof. Assume for contradiction that $f \neq g$. Then there exists an $x \in X$ such that $f(x) \neq g(x)$. Since Y is Hausdorff, there exist open neighbourhoods V_f and V_g of $f(x)$ and $g(x)$ respectively, such that $V_f \cap V_g = \emptyset$. Since f and g are continuous, there exist open neighbourhoods U_f and U_g of x such that $f(U_f) \subseteq V_f$ and $g(U_g) \subseteq V_g$. Define $U := U_f \cap U_g$. Then U is an open neighbourhood of x. Then we have $f(U) \cap g(U) \subseteq V_f \cap V_g$. Since A is dense, there exists a $y \in U \cap A$. Since $y \in A$, we have $f(y) = g(y)$. Then $f(y) \in f(U) \cap g(U)$, which implies $V_f \cap V_g \neq \emptyset$. This is a contradiction. Thus, $f = g$. ■

(b) Find an example which shows that this is not always true if Y is not Hausdorff.

Proof. Let $X = Y = \{a, b\}$ be equipped with the trivial topology, and let $A = \{a\}$. Then Y is not Hausdorff and A is dense in X. Let $f(x) = a$ and $g(x) = x$. Then $f|_A = g|_A$.
However, $f \neq g$. ■

Problem 4

Consider the following collection of subsets of $\mathbb{R}$:

$$\mathcal{B} = \{[a, \infty) \subseteq \mathbb{R}\colon a \in \mathbb{R}\}$$

(a) Show that $\mathcal{B}$ forms a basis for a topology $\mathcal{T}'$ on $\mathbb{R}$.

Proof. First, let $x \in \mathbb{R}$. Then $x \in [x, \infty) \in \mathcal{B}$.

Second, let $[a_1, \infty), [a_2, \infty) \in \mathcal{B}$, and define $a := \max(a_1, a_2)$. The intersection of these two sets is exactly $[a_1, \infty) \cap [a_2, \infty) = [a, \infty)$. Therefore, for any $x \in [a_1, \infty) \cap [a_2, \infty)$, we have $x \in [a, \infty) \in \mathcal{B}$, and clearly $[a, \infty) \subseteq [a_1, \infty) \cap [a_2, \infty)$. Thus, $\mathcal{B}$ forms a basis. ■

(b) Show that every continuous function $f\colon (\mathbb{R}, \mathcal{T}') \to (\mathbb{R}, \mathcal{T})$ is constant, where $\mathcal{T}$ denotes the usual topology on $\mathbb{R}$.

Proof. Assume for contradiction that there exist $x, y \in \mathbb{R}$ such that $f(x) \neq f(y)$. Then the singletons $\{f(x)\}$ and $\{f(y)\}$ are disjoint and closed in $(\mathbb{R}, \mathcal{T})$.
Since f is continuous, the preimage of a closed set is closed. All open sets in $(\mathbb{R}, \mathcal{T}')$ are unions of basis elements, so they are of the form $\emptyset$, $\mathbb{R}$, (a, ∞), or $[a, \infty)$ for some $a \in \mathbb{R}$. Thus, the closed sets in $(\mathbb{R}, \mathcal{T}')$ are their complements, which are of the form $\emptyset$, $\mathbb{R}$, $(-\infty, b]$, or $(-\infty, b)$ for some $b \in \mathbb{R}$.

Since $x \in f^{-1}(\{f(x)\})$, we have $f^{-1}(\{f(x)\}) \neq \emptyset$. The same holds for $f^{-1}(\{f(y)\})$. Notice that any non-empty closed set in $(\mathbb{R}, \mathcal{T}')$ must contain a set of the form $(-\infty, b)$ for some $b \in \mathbb{R}$. Therefore, there exist $b_x, b_y \in \mathbb{R}$ such that $(-\infty, b_x) \subseteq f^{-1}(\{f(x)\})$ and $(-\infty, b_y) \subseteq f^{-1}(\{f(y)\})$. Let $b = \min(b_x, b_y)$, then $(-\infty, b) = (-\infty, b_x) \cap (-\infty, b_y)$.
Then we have:

$$\emptyset \neq (-\infty, b) \subseteq f^{-1}(\{f(x)\}) \cap f^{-1}(\{f(y)\}) = f^{-1}(\{f(x)\} \cap \{f(y)\}) = f^{-1}(\emptyset) = \emptyset$$

This is a contradiction. Thus, f must be constant. ■

Problem 5

Let $A = S^1 \cap \{(x, y) \in \mathbb{R}^2 \mid y \geq 0\}$, $B = \{(x, y) \in \mathbb{R}^2 \mid x \in [-1, 1], y = 0\}$, and let $X = A \cup B$, endowed with the subspace topology from $\mathbb{R}^2$. Show that X is homeomorphic to S^1.

Proof. Define the bijection $f\colon S^1 \to X$ as follows:

$$f(x, y) = \begin{cases} (x, y), & \text{if } y \geq 0, \\ (x, 0), & \text{if } y \leq 0. \end{cases}$$

The restriction $f|_{\{y \leq 0\}}$ is a projection map, and is therefore continuous. Similarly, the restriction $f|_{\{y \geq 0\}}$ is the identity map, which is also continuous. Because these restrictions are defined on closed subsets of S^1, the pasting lemma implies that f is continuous.

Since S^1 is a closed and bounded subset of $\mathbb{R}^2$, it is compact. Furthermore, X is Hausdorff as it is a subspace of $\mathbb{R}^2$. Because f is a continuous bijection from a compact space to a Hausdorff space, it is a homeomorphism. ■

Problem 6

Define an equivalence relation on $\mathbb{R}$ as follows:

$$x \sim y \iff x - y \in \mathbb{Z}.$$

Write $\mathbb{R}/\mathbb{Z}$ for the quotient $\mathbb{R}/\sim$. Show that $\mathbb{R}/\mathbb{Z}$ is homeomorphic to S^1.

Proof. Define $f\colon \mathbb{R} \to S^1$ by $f(x) = (\cos(2\pi x), \sin(2\pi x))$. The map f is continuous since its component functions are continuous, and it is trivially surjective.

To show that f is a quotient map, let $C \subseteq S^1$ be a subset such that $f^{-1}(C)$ is closed in $\mathbb{R}$. We must show that C is closed in S^1. Let $(y_n)_{n \in \mathbb{N}}$ be a sequence in C that converges to $y \in S^1$. Because f is periodic with period 1, for each $n \in \mathbb{N}$ we can choose an $x_n \in [0, 1]$ such that $f(x_n) = y_n$. Thus, $(x_n)_{n \in \mathbb{N}}$ is a sequence in $f^{-1}(C) \cap [0, 1]$.

Because $f^{-1}(C)$ is closed by assumption and $[0,1]$ is closed and bounded, their intersection $f^{-1}(C) \cap [0,1]$ is a closed and bounded subset of $\mathbb{R}$. Therefore, it is compact, and hence sequentially compact. This guarantees the existence of a subsequence $(x_{n_k})_{k\in\mathbb{N}}$ that converges to some $x \in f^{-1}(C) \cap [0,1]$.

Since f is continuous, $y_{n_k} = f(x_{n_k}) \to f(x)$ as $k \to \infty$. Furthermore, $f(x) \in C$ because $x \in f^{-1}(C)$. Because (y_{n_k}) is a subsequence of the convergent sequence (y_n), it must converge to the same limit. Thus, $y = f(x) \in C$. This proves that C is closed, which means f is a quotient map.

Now, let $x, y \in \mathbb{R}$. We have:

$$\begin{aligned} f(x) = f(y) &\iff (\cos(2\pi x), \sin(2\pi x)) = (\cos(2\pi y), \sin(2\pi y)) \\ &\iff x = y + k \text{ for some } k \in \mathbb{Z} \\ &\iff x - y \in \mathbb{Z} \\ &\iff x \sim y. \end{aligned}$$

This shows that the equivalence relation induced by f is exactly the one given by $\sim$. Therefore, f induces a homeomorphism from the quotient space $\mathbb{R}/\mathbb{Z}$ to S^1. ■

Problem 7

Let $X = \prod_{n\in\mathbb{N}}\{-1,1\} = \{-1,1\}^{\mathbb{N}}$ with the product topology (where $\{-1,1\}$ has the discrete topology).

(a) For $0 < q < 1$, show that $f\colon X \to \mathbb{R}$,

$$f((x_n)_{n\in\mathbb{N}}) = \sum_{n=1}^{\infty} x_n q^n$$

is continuous.

Proof. Let $V = B(x,r)$ be a basis element in $\mathbb{R}$, and let $(x_n)_{n\in\mathbb{N}} \in f^{-1}(V)$.
Define $\varepsilon := \frac{1}{2}\big(r - |x - f((x_n)_{n\in\mathbb{N}})|\big) > 0$. Then let $N \in \mathbb{N}$ be such that $\sum_{n=N+1}^{\infty} q^n < \varepsilon$, and define

$$U := \bigcap_{n=1}^{N} \pi_n^{-1}(\{x_n\}).$$

Let $(y_n)_{n\in\mathbb{N}} \in U$. By construction, $x_n = y_n$ for all $n \le N$. Then we have:

$$\begin{aligned} |f((y_n)_{n\in\mathbb{N}}) - x| &\le |f((y_n)_{n\in\mathbb{N}}) - f((x_n)_{n\in\mathbb{N}})| + |f((x_n)_{n\in\mathbb{N}}) - x| \\ &\le \left|\sum_{n=N+1}^{\infty} y_n q^n\right| + \left|\sum_{n=N+1}^{\infty} x_n q^n\right| + r - 2\varepsilon \\ &\le 2\sum_{n=N+1}^{\infty} q^n + r - 2\varepsilon \\ &< 2\varepsilon + r - 2\varepsilon = r. \end{aligned}$$

Therefore, $f((y_n)_{n\in\mathbb{N}}) \in V$, which implies $U \subseteq f^{-1}(V)$. Thus, $f^{-1}(V)$ is open, and hence f is continuous. ■

(b) For $0 < q < \frac{1}{2}$, show that f is injective.

Proof. Let $(x_n)_{n\in\mathbb{N}}, (y_n)_{n\in\mathbb{N}} \in X$ be such that $f((x_n)_{n\in\mathbb{N}}) = f((y_n)_{n\in\mathbb{N}})$. We proceed by induction on k to show that the first k elements of $(x_n)_{n\in\mathbb{N}}$ and $(y_n)_{n\in\mathbb{N}}$ are equal. For the base case $k = 1$, assume for contradiction that $x_1 \neq y_1$. Then we have:

$$
\begin{aligned}
0 &= \left|\sum_{n=1}^{\infty} x_n q^n - \sum_{n=1}^{\infty} y_n q^n\right| \\
&= \left|\sum_{n=1}^{\infty} (x_n - y_n) q^n\right| \\
&\geq |x_1 - y_1| q - \left|\sum_{n=2}^{\infty} (x_n - y_n) q^n\right| && \text{(inverse triangle inequality)} \\
&= 2q - q\left|\sum_{n=1}^{\infty} (x_{n+1} - y_{n+1}) q^n\right| && \text{(since } x_1 \neq y_1\text{)} \\
&\geq 2q - q\sum_{n=1}^{\infty} |x_{n+1} - y_{n+1}| q^n && \text{(triangle inequality)} \\
&\geq 2q - 2q\sum_{n=1}^{\infty} q^n \\
&= 2q - \frac{2q^2}{1-q} > 0, && \text{(since } 0 < q < \tfrac{1}{2}\text{)}
\end{aligned}
$$

which is a contradiction. Thus, $x_1 = y_1$.

Assume that for some $k \geq 1$, $x_n = y_n$ for all $n \leq k$. We want to show that $x_{k+1} = y_{k+1}$.

We still have that:

$$
\begin{aligned}
0 &= \left|\sum_{n=1}^{\infty} x_n q^n - \sum_{n=1}^{\infty} y_n q^n\right| \\
&= \left|\sum_{n=k+1}^{\infty} x_n q^n - \sum_{n=k+1}^{\infty} y_n q^n\right| && \text{(induction hypothesis)} \\
&= q^k \left|\sum_{n=1}^{\infty} x_{n+k} q^n - \sum_{n=1}^{\infty} y_{n+k} q^n\right| \\
&\Leftrightarrow f((x_{n+k})_{n\in\mathbb{N}}) = f((y_{n+k})_{n\in\mathbb{N}}) && \text{(definition of } f\text{)} \\
&\Rightarrow x_{k+1} = y_{k+1}. && \text{(base case for } k = 1\text{)}
\end{aligned}
$$

By induction, $x_n = y_n$ for all $n \in \mathbb{N}$, and therefore $(x_n)_{n\in\mathbb{N}} = (y_n)_{n\in\mathbb{N}}$. Thus, f is injective. ■

(c) For $0 < q < \frac{1}{2}$, show that f is actually an embedding, by showing that $f\colon X \to f(X)$ is open.

Proof. let ■